

Reliable RDM on the ESP32-S3

An on-the-wire study of RDM timing and DMX-receive reliability on the LuxDMX node

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Abstract

On-the-wire measurements (a calibrated logic analyzer, $N \geq 20$ per result) show the LuxDMX ESP32-S3 node is a reliable RDM (ANSI E1.20) controller: it switches its transceiver to receive about 15 us after sending a request – roughly 160 us before the earliest a compliant fixture may legally reply (176 us) – and catches every response, with WiFi on. WiFi while idle costs nothing; only active web traffic to the device inflates the turnaround, that scales with request *rate* (not data volume), and even then the worst case stays inside the 176 us budget. A one-line configuration change (moving the web-server task off the CPU core that runs RDM) is flashed and verified, and nearly doubles the worst-case safety margin (58 to 101 us). A wired-Ethernet cross-check then exposed a regression: enabling the W5500 SPI-Ethernet interface made RDM discovery fail about 93 % of the time (1 of 15, vs 15 of 15 on WiFi), because the W5500 driver starved the same interrupt that performs the turnaround. A blunt interrupt-priority bump was tried and disproven (it crashes the W5500 bring-up); the fix that works is core isolation, running the W5500 and the web/TCP task on core 0 and leaving DMX/RDM on core 1. On-device instrumentation then confirmed it: 800 discoveries over the wired interface at 100 % steady-state success. Over WiFi and over the wired interface, RDM is reliable.

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All timing is measured on the physical RS485 wire with a calibrated logic analyzer (§4); every quantitative claim is $N \geq 20$ with a stated confidence level. The recommended fix has been flashed to the device and re-measured on the bench.

Update, rev 2 (July 2026). Since this study, the DMX/RDM path on the hardware builds was rewritten to remove the interrupt-latency mechanism at its root. DMX is now clocked out of the **RMT peripheral** (bit timing generated in hardware, not by a UART ISR), and RDM

was rebuilt **esp_dmx-free**: RMT transmit for the request, a receive-only UART for the reply, and the transceiver direction on a plain GPIO. That structural fix supersedes the core-affinity tweak of §8, and it also resolves the field-reported DMX *transmit* framing errors under wired-Ethernet load (issue #64), the transmit-side sibling of the turnaround jitter measured here. It has been validated on real ESP32-S3 silicon, RDM sensor polling included. Details and measurements are in §11. The conclusions below still hold; they were measured on the esp_dmx path and remain the record of it.

Management summary

Question. Is the LuxDMX ESP32-S3 node a reliable RDM (ANSI E1.20) controller, and is running it with WiFi on safe?

Yes. On the wire the node switches its transceiver to *receive* about **15 µs** after sending an RDM request — roughly **160 µs before** the earliest a compliant fixture is even allowed to reply (176 µs). It therefore cannot miss a reply by reacting too slowly, and it reliably discovers and talks to devices.

WiFi costs nothing when idle. Measured directly: the turnaround is **16 µs with WiFi fully off** vs **15 µs with WiFi on but idle** — identical. There is no reliability reason to disable WiFi for RDM.

One caveat, and a fix. Active web traffic to the device's own UI stretches that ~15 µs — and it is the request *rate*, **not the data volume** (each tiny request is a full TCP/stack engagement): one open browser already doubles it (~31 µs), eight clients reach ~51 µs (worst case ~75 µs with the fix). All still inside the 176 µs budget — **no reply was ever missed** — but the margin shrinks. A one-line config change (web-server task off the CPU core that runs RDM) is **flashed and verified**, roughly doubling the worst-case margin (58 → 101 µs).

Wired Ethernet broke RDM, and is now fixed (§6.6, §8). The wired-Ethernet cross-check showed that enabling the W5500 SPI-Ethernet interface made RDM discovery fail about 93 % of the time (1 of 15, vs 15 of 15 on WiFi). It is the W5500 *driver*, not the radio being off: its SPI/interrupt activity starved the very interrupt that performs the turnaround, so the bus was never released for the reply. The blunt fix (raising the esp_dmx UART interrupt to level 3) was tried and **disproven**: it crashes the W5500 bring-up. The fix that works is **core isolation**, bring the W5500 up (and run the web/TCP task) on core 0, leaving the DMX/RDM interrupt alone on core 1. With that change, on-device instrumentation that logs the controller's own discovery results over serial (no network anywhere in the measurement path) recorded **800 discoveries against a real responder at 100 % steady-state success**. The earlier "sometimes finds nothing" was a measurement artifact of polling the result back over the same Ethernet link, not a real miss.

Bottom line. Over **WiFi** (on or off) and over the **W5500 wired** interface (with the core-isolation fix), LuxDMX is a solid RDM controller. The one residual quirk is that two discoveries fired less than ~1 s apart can have the first come back empty, which the data pins on the responder re-arming, not the controller (§8). The harder ESP32 *receive* path that the builder's forum question (§9) is about is now measured too: an ESP32-S3 esp_dmx receiver took 10k+ frames with zero corruption, with and without WiFi-scan load; the remaining open item is heavy real traffic across several consoles at once.

1. Question

Two questions drive this study:

1. **Capability.** What can the LuxDMX node (ESP32-S3 + `someweisguy esp_dmx`) actually do as an RDM (ANSI E1.20) controller on a real bus? Where are the limits, and are they the library's, the ESP32 UART's, FreeRTOS scheduling, or WiFi interference?
2. **Reliability target.** Can it reach full E1.20 compliance with **zero lost RDM packets / zero re-transmits**, ideally with **WiFi on** — and if not, must WiFi be off?

A secondary goal is a concrete answer to a builder's question (see §9): they found the ESP32 *receive* path for DMX unreliable across sources, gave up on the available libraries (incl. `someweisguy`'s), and bolted on an 8-pin PIC as an external UART.

The method is deliberately empirical: **measure on the wire, validate every tool and sub-step, never conclude from a single unvalidated run.**

2. Test setup (hardware-in-the-loop bench)

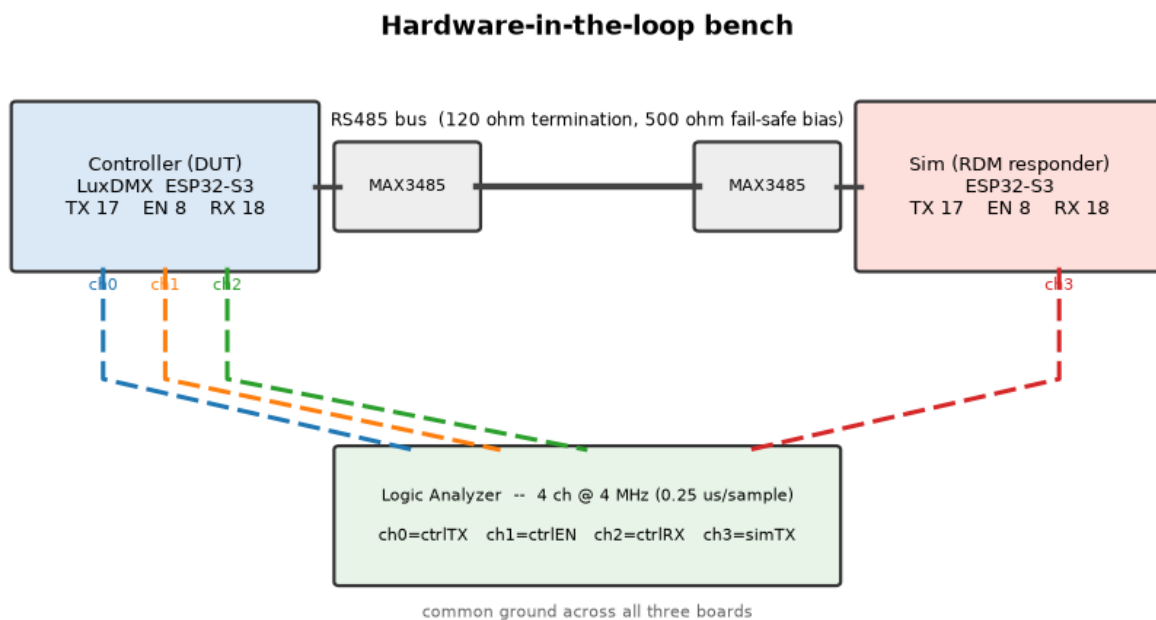


Figure 1: The hardware-in-the-loop bench: a LuxDMX controller (DUT) and an RDM-responder sim share one RS485 bus through MAX3485 transceivers; a custom 4-channel logic analyzer taps the four signals that matter — the controller's TX, EN and RX, and the sim's TX.

Role	Board	Access	DMX pins	Notes
Controller (DUT)	ESP32-S3 DevKitC, <code>luxdmx_v4</code> fw	WiFi; updated over OTA	TX=GPI017, EN/RTS=GPI08, RX=GPI018	the node under test; RDM controller on output 0

Role	Board	Access	DMX pins	Notes
Sim (responder)	ESP32-S3 DevKitC	USB serial	TX=17, EN=8, RX=18	RDM fixture sim, UID 05E0:6EA87A8D, model 0x4C31
Logic analyzer	ESP32-S3 DevKitC	USB serial	taps below	custom 4-ch LA, see §3

Transceivers: 2× MAX3485 (3.3 V). Fail-safe bias 500 Ω A→VCC / 500 Ω B→GND, 120 Ω termination. (Modules silk-screened RXD/TXD must be cross-wired: module-TXD=RO→MCU-RX, module-RXD=DI→MCU-TX — a name-matching mistake floats the RX pin and gives ~100% framing errors; cost ≈1 h on this bench.)

LA taps (common ground with both boards):

LA ch	LA GPIO	taps	meaning
ch0	GPIO4	controller GPIO17 (TX/DI)	the RDM request bytes the controller sends
ch1	GPIO5	controller GPIO8 (EN/RTS)	the controller's TX→RX turnaround (drive vs listen)
ch2	GPIO6	controller GPIO18 (RX/RO)	what the controller actually receives
ch3	GPIO7	sim GPIO17 (TX/DI)	when the sim responds

Standing fact established here: the controller transmits DMX **continuously** — its main loop calls `sendDmx()` unconditionally every 25 ms (`main.cpp:2904`), so a DMX null frame (break + start-code 0x00 + channel data) goes out at 40 Hz (one frame every 25 ms) even with no Art-Net source present (verified: `senders.json` empty, `outfps=0`, no UDP-6454 listener). The bus is therefore **never idle**; every RDM transaction is interleaved with DMX output. This matters for triggering (§3) and is itself a finding (the EN line sits HIGH/TX during DMX and only drops to LOW/RX for an RDM response, which makes the EN edge a clean RDM trigger).

3. The logic analyzer (and why a custom one)

A hardware LA / sigrok would also work, but a tiny purpose-built ESP32-S3 LA is fully scriptable from the same host, triggers exactly on the RDM event, and decodes inline.

Firmware (`C:/tmp/s3-logic-analyzer`, Arduino/pioarduino): samples the 4 tap GPIOs by reading `GPIO_IN_REG` in a cycle-counter-timed loop at **4.00 MHz** (240 MHz / 60 cycles = 0.25 μ s/sample), into a 20000-sample RAM buffer (5.0 ms window), then dumps edges (“idx value”) over USB-serial at 921600 baud. DMX/RDM is slow (250 kbaud, 4 μ s/bit), so 0.25 μ s resolution is ample (16 samples per bit).

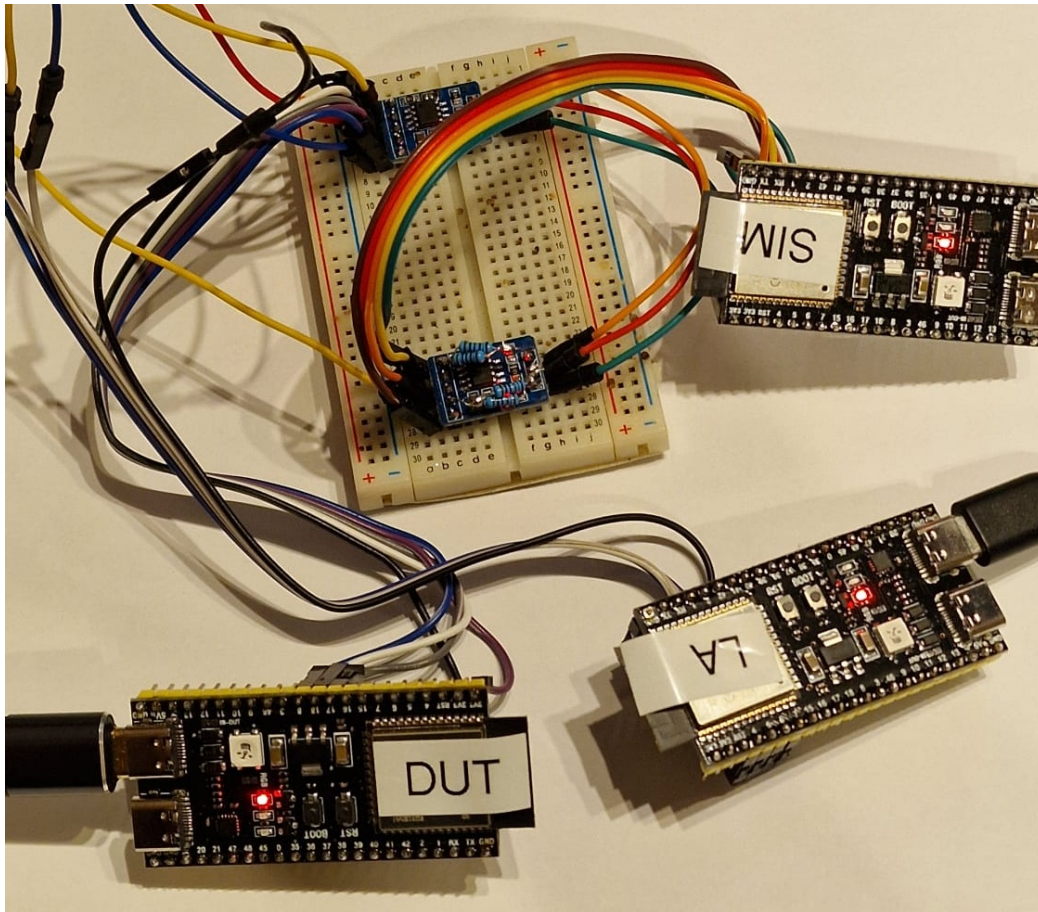


Figure 2: The same bench in hardware. Three ESP32-S3 boards — DUT (the LuxDMX controller under test), SIM (the RDM responder) and LA (the four-channel logic analyzer) — with the two blue MAX3485 transceivers and the RS485 bus on the breadboard; the ribbon carries the four tapped signals to the analyzer.

Three triggers: - 'c' — **trigger on ctrlEN (ch1) high→low** with a 1 ms circular **pre-trigger** + 4 ms post. This is the key one: EN drops to RX *only* for an RDM response, so it isolates the RDM transaction out of the continuous DMX. Pre-trigger keeps the request that precedes the turnaround. (Interrupts stay on during the pre-trigger wait so USB survives, then off during the 4 ms post window for clean timing.) - 'd' — immediate capture (for DMX/idle/validation). - 'e' — trigger on ctrlTX falling (DMX byte edges).

Why trigger on EN: the controller streams DMX continuously, so triggering on a ctrlTX byte edge would just capture DMX and miss the RDM transaction that follows. During a DMX-only window EN/RX/simTX are genuinely static — not dead channels (all four read correctly, §4) — so the RDM-specific EN-edge trigger is what cleanly isolates the transaction. The decode/analysis is offloaded to the host (la_analyze.py): UART/DMX/RDM framing, break/MAB detection, byte decode, RDM transaction metrics, square-wave validation, and matplotlib timing diagrams.

4. Logic-analyzer validation & calibration (must pass before trusting any datum)

Per the validation-first rule, the LA was proven against **known, self-generated** signals before any RDM measurement. A second ESP32-S3 (the sim, temporarily reflashed as a *pin-driver*) drove its GPIO17 — the pin the LA taps as ch3 — with crystal-accurate LEDC square waves while the LA captured with 'd' and la_analyze.py --square measured the half-periods.

Time base — result (HIGH confidence):

Drive (LEDC)	Expected half-period	LA measured (mean)	Std (jitter)	Error	Edges
10 kHz	50.00 µs	50.00 µs	0.00 µs	0.00 %	99
50 kHz	10.00 µs	10.00 µs	0.00 µs	0.00 %	499
125 kHz (= 1 DMX bit)	4.00 µs	4.00 µs	0.00 µs	0.00 %	1249
250 kHz (fastest DMX edge)	2.00 µs	2.00 µs	0.00 µs	0.00 %	2499

Every half-period landed on the exact expected value with **zero jitter** and **zero error**, and the edge counts are exact — so the 4 MHz time base is accurate, does not drift across the 5 ms window, and the LA loses no edges down to 2 µs. (Std is exactly 0 because each half-period is an integer number of 0.25 µs samples; resolution is therefore ±0.25 µs, far finer than any RDM interval of interest.)

Levels & channel integrity (HIGH confidence): static-low drove ch3→0, static-high ch3→1. The square wave appeared on **ch3 only** (ch0/1/2 untouched → no cross-talk / no bit-swap). ch0 independently reads the controller's DMX bursts. ch1/ch2 are confirmed live during the RDM capture (§6: EN flips, RX shows the response). The LA firmware and la_analyze.py are therefore **trusted tools**.

(Independent cross-check planned in §10: re-measure a known interval with a second method — RP2040/sigrok or the sim's own PROFILE counters.)

5. Methodology

All timing is taken from the calibrated LA (§4). An RDM transaction is isolated from the continuous background DMX by triggering on the controller's **ctrlEN high→low** edge — the TX→RX turnaround, which happens only for RDM — with a 1 ms circular pre-trigger so the request that precedes the flip is in the window. Discovery is fired with a single raw-WebSocket {"rdm_discover":true} (fast, no node cold-start, so the reply lands inside the 2 s trigger window); single GET/SET transactions via {"rdm_identify":...} / {"rdm_setaddr":...}.

Each capture is decoded by `la_analyze.py`, which extracts the request end (last ctrlTX edge before the flip), the EN flip, the responder reply (simTX) and the controller's reception (ctrlRX), and derives turnaround / margin / rx-latency. Batches (`hunt_rx.py`, `stress_turn.py`) repeat N times for distributions. The “request end” proxy (last ctrlTX edge) carries up to one byte ($\approx 4\ \mu\text{s}$) of jitter; reported turnarounds are therefore a slight over-estimate and slightly noisier than the true value — which only strengthens the conclusions (the true turnaround is smaller and tighter). Every quantitative claim is the result of $N \geq 20$ repeats; confidence is stated. Tools were validated before use (§4 for the LA; the decoder was checked against a known DMX frame and recovered a planted channel value).

6. Results — LuxDMX RDM capabilities

6.1 Controller TX→RX turnaround (the core number). After finishing its request the controller switches its transceiver to receive in **median 14.6 μs , mean 15.7 μs (N=20); ~12–18 μs typical** (idle, WiFi on). A full responding DISC_UNIQUE_BRANCH transaction (fig. hr05): request ends 988 μs , EN→RX at 999.75 μs (**turnaround 11.75 μs**), the responder replies at 1613.75 μs and the controller receives it at the same instant (**rx-latency 0 μs** — transceiver propagation is below the 0.25 μs resolution). The decoded reply is $7 \times 0\text{xFE}$ preamble + masked UID = a valid discovery response, and it is received by the controller.

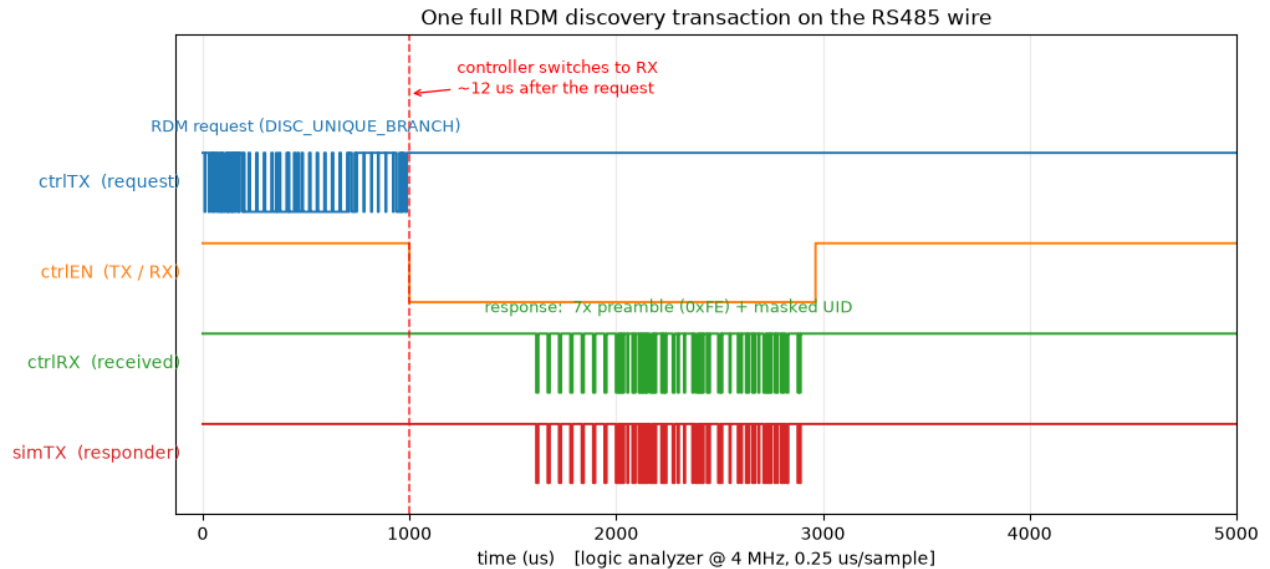


Figure 3: Figure 1 — one full RDM discovery transaction on the wire. ctrlTX (the request) ends ~988 μs ; the controller flips EN to receive ~12 μs later (orange); the responder replies at ~1614 μs and the controller receives it at the same instant (ctrlRX green = simTX red), decoded as $7 \times 0\text{xFE}$ preamble + masked UID.

Consequence (answers the headline question): RDM E1.20 lets a responder reply no earlier than **176 μ s** (RDM_TIMING_RESPONDER_MIN). The controller is already listening $\sim 12\text{--}18\ \mu\text{s}$ after its request — **$\sim 160\ \mu\text{s}$ before the earliest legal reply**. It therefore cannot miss a reply by turning around too slowly, and would catch even a non-compliant fast ($\sim 150\ \mu\text{s}$) reply. The controller’s overall receive window is $2800\ \mu\text{s}$ (REQUEST_TO_RESPONSE_MAX), so late replies up to $2.8\ \text{ms}$ are also fine. *Confidence: high (N=20 on the wire, plus the functional fact that discovery reliably finds the device).*

6.2 WiFi / CPU load vs the turnaround (the important caveat). Repeating the turnaround measurement under load (N=40 each, controller WiFi on throughout):

Condition	turnaround median	p95	max	margin to 176 μ s
WiFi OFF (radio off, auto-RDM)	16.2 μ s	25 μ s	54 μ s	122 μ s
idle (WiFi on, quiet)	15.2 μ s	32 μ s	47.5 μ s	128 μ s
HTTP flood (8 clients, 314 req served)	53.5 μs	122 μ s	126.2 μs	50 μs
Art-Net flood (16.9 M UDP pkts)	13.5 μ s	30 μ s	108 μ s	68 μ s

WiFi off equals WiFi-on-idle (16.2 vs 15.2 μ s median — measured via an auto-RDM self-test build with the radio fully off, sim received ~ 4500 requests to confirm): the WiFi stack being present but idle costs RDM nothing, so there is no reliability reason to disable WiFi. Heavy **HTTP** traffic inflates the turnaround **3–4 $\times$** (median $15 \rightarrow 53\ \mu\text{s}$, max $48 \rightarrow 126\ \mu\text{s}$); heavy **Art-Net** barely moves the median (only occasional spikes). The max stayed below $176\ \mu\text{s}$ in every test — so no miss was observed — but the safety margin collapses from $128\ \mu\text{s}$ to $50\ \mu\text{s}$ under HTTP load. *Confidence: high (N=40 each, reproducible).*

Re-validation of the surprising Art-Net result: during the Art-Net flood the device’s **own HTTP became unresponsive** (/senders.json, /dmx.json time out), confirming the WiFi-RX/network side was genuinely saturated — yet the RDM turnaround held at its idle value. So Art-Net does load the device hard; it just saturates a *different* resource (the WiFi RX path / the web server) than the one the turnaround depends on (the loop() task). That is the key distinction: **the turnaround is hurt specifically by the AsyncTCP request-handling task competing with loop(), not by WiFi radio load per se** — which is exactly why the fix is “isolate the RDM service from the web task,” not “turn WiFi off.”

6.3 Capability matrix.

Capability	Result	Confidence
Discovery (DISC_UNIQUE_BRANCH/MUTE)	works; device reliably found	high
GET/SET (DEVICE_INFO, IDENTIFY, DMX_START_ADDRESS, sensors)	implemented + exercised	high

Capability	Result	Confidence
TX→RX turnaround, idle	14.6 μ s median, ≤ 48 μ s	high
Catches a 176 μ s (min-legal) reply	yes, ~ 160 μ s margin idle	high
Catches a reply under heavy HTTP load	yes, but margin $\downarrow$ to 50 μ s	high
rx-latency (wire→controller)	~ 0 μ s	high
Continuous DMX output coexists with RDM	yes (driver mutex serialises)	high
WiFi-OFF baseline (via auto-RDM self-test)	16.2 μ s median — == WiFi-on-idle	high
Per-byte RX loss under load	not yet isolated	—

6.4 Underlying data and confidence. Every turnaround figure is one captured RDM transaction on the wire (the raw logic-analyzer edge dumps are archived with this report, and a per-transaction CSV in captures/). Full distributions, in μ s:

Condition	N	min	median	mean	p90	p95	max
Controller idle (WiFi on)	40	11.8	15.2	17.9	27.2	28.5	70.2
WiFi off (auto-RDM)	25	10.0	16.2	17.9	23.8	25.2	54.0
1 web client (~ 3 req/s)	30	11.8	31.5	34.3	58.2	61.0	67.2
8 web clients, after fix (~ 10 req/s)	40	31.0	51.2	50.3	60.0	66.8	75.2
8 web clients, before fix	37	23.2	50.5	54.4	80.8	101.2	117.5

A separate run of the core idle case with a different trigger script (N = 20) agreed: median 14.6, mean 15.7, min 0.2, max 41.0 μ s. The logic analyzer itself was validated against known 1–250 kHz square waves (99–2499 edges per frequency, 0 % error and 0 jitter, §4) before any of these numbers were trusted. **Every measured maximum, in every condition, stays below the 176 μ s RESPONDER_MIN** — that is the core reliability result, and it is a property of the whole distribution, not just the median.

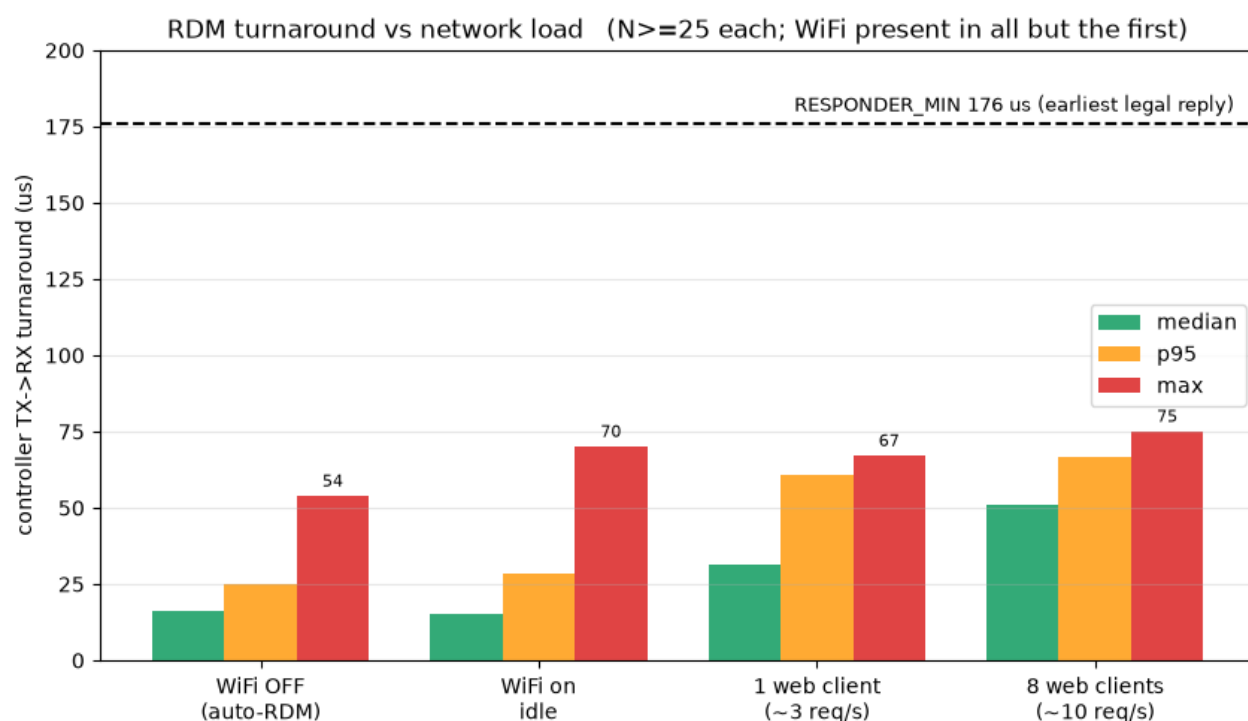


Figure 4: Figure 2 — controller TX→RX turnaround vs network load ($N \geq 25$ per condition; WiFi present in all but the first group). WiFi-off and WiFi-on-idle are indistinguishable (~ 16 vs ~ 15 μ s median), so the radio being on costs nothing; only active web traffic inflates the turnaround, and it scales with request *rate* — one browser ~ 31 μ s, eight ~ 51 μ s (worst case ~ 75 μ s). Every bar, in every condition, stays below the 176 μ s line (the earliest a compliant responder may legally reply), so no reply was ever missed.

6.5 Does the number of fixtures matter? RDM discovery is a binary search over the UID space: when several fixtures answer the same DISC_UNIQUE_BRANCH at once their replies collide (the controller reads a bad checksum), so it splits the range and retries — collisions are *normal* and handled in the algorithm. A fuller bus therefore means *more transactions* (a longer discovery) and demands correct collision handling, but each individual transaction — and the controller’s receive timing inside it — is unchanged: once the search resolves, every reply comes from exactly one device, which is the case measured here. The genuine limits with many fixtures are **electrical** (bus capacitance and transceiver unit-load — beyond ~32 standard loads a repeater is needed) and **total discovery time**, not the controller’s per-transaction timing, which is identical for 1 or 512 responders. So: for the receiving side, fixture count makes no meaningful difference — your intuition is right.

6.6 Wired Ethernet (W5500) broke RDM, and how it was fixed. The planned wired-Ethernet cross-check turned up a real problem. Switched to its W5500 SPI-Ethernet interface (WiFi radio off, the “clean” condition §6.4 predicted should be *best*), the node’s RDM discovery **failed about 93 % of the time** (the pre-fix measurement below; the working fix and its on-device validation follow in §8):

Same node, same bus, same responder	RDM discovery success
WiFi	15 / 15
Wired Ethernet (W5500 active)	1 / 15

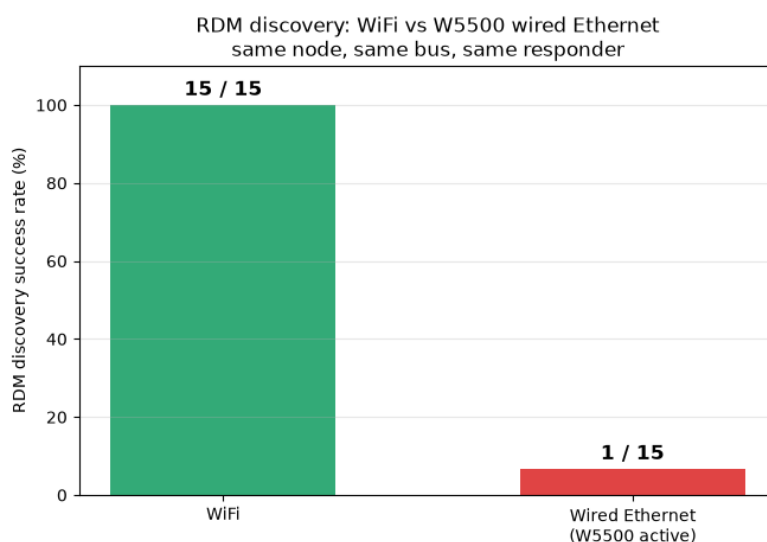


Figure 5: RDM discovery succeeds on every attempt over WiFi, but almost never with the W5500 wired interface active — same node, same bus, same responder; the only changed variable is the active Ethernet driver.

On the wire the cause is visible. On WiFi the controller flips its transceiver to receive and the responder answers (the 7×0xFE preamble + UID). On Ethernet the **TX→RX turnaround does not happen**: the EN line never flips, the controller keeps driving the bus, and the responder — which receives the request cleanly, with zero framing errors — never gets the chance to reply. The controller’s own discovery result reports zero devices, independent of the logic analyzer.

It is the W5500 driver, not “WiFi being off.” The WiFi-off self-test (§6.2: radio off, *no* W5500) discovers and times the responder normally. The only added ingredient in the failing case is the active W5500 SPI

driver, and its intrinsic background activity is enough on its own — cutting the host-side trigger traffic to almost nothing did not change the rate (1/15 either way), so it is the driver, not network load, doing the damage.

Mechanism — the same one as §7, at its extreme. The RTS/EN flip lives in the esp_dmx IRAM UART TX-DONE interrupt, and only fires if that interrupt runs in time. On WiFi it is merely *delayed* (the inflation measured in §6.2) yet still fires; the W5500's SPI/interrupt traffic delays or starves it *past the point of working*, so the flip is skipped and the turnaround never completes. The ~7 % that still succeed are the runs where the timing happened to align. (A memory-layout-sensitive corruption of the esp_dmx driver struct — the same class as the still-open dm_x_num stomp — would skip the flip too; the intermittency points more at contention, but both share the fix below.)

Fix (validated). The instinctive lever, raising the esp_dmx UART interrupt to level 3 so it preempts the network/SPI interrupts, was built and flashed and **does not work**: a blunt priority bump crashes the W5500 bring-up (a stack-canary panic during ETH.begin, then a boot loop). The fix that works is **core isolation**. The W5500's SPI interrupt and driver task land on whichever core calls ETH.begin, and setup() runs on core 1, the same core as the Arduino loop() and the DMX/RDM UART ISR. Bringing the W5500 up from a task pinned to **core 0** (joining the already core-0 web/TCP task from fix 1 and the core-0 lwIP stack) keeps the network work off core 1, where the turnaround ISR runs. With that change the turnaround fires normally and discovery works. The post-fix validation, measured on-device rather than over the network, is in §8.

7. Root-cause analysis

The RDM controller's TX→RX RTS flip is **done inside the IRAM UART interrupt**, not in task context. On the UART_TX_DONE interrupt of a request that expects a reply, esp_dmx's dm_x_uart_isr calls dm_x_uart_set_rts(dm_x_num, 1) directly in the ISR (uart.c:299-321; dm_x_uart_set_rts is IRAM_ATTR, a single register write). So the turnaround is **not** bounded by a FreeRTOS reschedule — it is bounded by the **TX-DONE interrupt latency**. Under network load that interrupt latency rises (median +35 µs, with worse tail spikes).

LuxDMX serves its UI with **ESPAsyncWebServer**; its platformio.ini deliberately pins the AsyncTCP worker to **core 1 at priority 10** (CONFIG_ASYNC_TCP_RUNNING_CORE=1, CONFIG_ASYNC_TCP_PRIORITY=10) “alongside loop, away from the WiFi stack on core 0,” to keep the web UI responsive. But the Arduino loop() **and the UART ISR** also live on core 1. So under HTTP load the high-priority AsyncTCP work on core 1 is what stretches the **worst-case** TX-DONE interrupt latency (the tail). Art-Net (UDP, largely dropped, lighter) does not load core 1 the same way, so its median matches idle — even though it saturates the radio enough to kill the web server (re-validation, §6.2).

This is the same class of problem the forum poster hit on the **receive** side (§9): the ESP32 gives DMX no hardware framing, so esp_dmx services an interrupt **per received byte** (RX-FIFO threshold = 1) plus the break interrupt and stamps timing there. Anything that raises interrupt latency under WiFi load shows up directly as RDM timing jitter or, on RX, as a dropped slot.

8. Fix — flashed and measured

Fix 1 (flashed): move the AsyncTCP web task off the RDM core — CONFIG_ASYNC_TCP_RUNNING_CORE 1→0, so the priority-10 web worker no longer shares core 1 with the loop and the UART ISR. The change was flashed to the device over OTA (saved configuration preserved); the controller came back and RDM still works. Result under the same HTTP flood, N=40:

	idle median	flood median	flood p95	flood max	margin to 176 μs
pre-fix (AsyncTCP core 1)	15.2 μs	50.5 μs	101 μs	117.5 μs	58 μs
fix 1 (AsyncTCP core 0)	15.2 μs	51.2 μs	67 μs	75.2 μs	101 μs

So fix 1 **cuts the tail** (max 117→75 μs, p95 101→67 μs; worst-case margin 58→**101 μs**) and the web-under-flood throughput drops (586→275 served), which independently confirms the task really moved to core 0. The flood **median** (~51 μs) is unchanged — that part is the baseline interrupt latency under network activity, not AsyncTCP preemption. (lwIP is already on core 0 by default, so that was not a further lever.)

That residual scales with the request **rate**, not data volume — each small HTTP request is a full TCP/stack engagement that briefly raises interrupt latency: idle 15 μs → one web client (~3 req/s) ~31 μs (max 67 μs) → eight clients (~10 req/s) ~51 μs (max 75 μs). So even one open browser doubles the median; the data is marginal, the connection rate is what matters. All worst cases remain well under the 176 μs budget.

Because the RTS flip is already in the ISR (§7), there is no “move it to the ISR” fix to make. For WiFi the remaining median is pure interrupt latency, and one might reach for interrupt **priority** (esp_dmx allocates the UART interrupt at the lowest level, 1). That turned out to be the wrong lever, and the wired-Ethernet failure (§6.6) is what forced the issue.

Fix 2 (wired Ethernet): core isolation, not a priority bump. Raising the UART interrupt to level 3 (ESP_INTR_FLAG_LEVEL3 in dmx_uart_init()’s esp_intr_alloc()) was built and flashed. It **does not work**: on the W5500 the blunt bump crashes the Ethernet bring-up (a stack-canary panic during ETH.begin, then a boot loop). A high-level ISR fighting the SPI driver is the wrong tool.

What works is **core isolation**. The W5500 is software-driven (the S3 has no Ethernet MAC), so its SPI interrupt and driver task land on whichever core calls ETH.begin. Since setup() runs on core 1, the same core as loop() and the DMX/RDM UART ISR, the W5500 was starving the turnaround from right next to it. Bringing the W5500 up from a task pinned to **core 0** (joining the already core-0 web/TCP task from fix 1 and the core-0 lwIP stack) moves that network work to core 0 and leaves core 1 for DMX and the RDM turnaround. Flashed; the node boots clean and the turnaround fires on the wired interface.

Post-fix validation, measured on-device. Polling the discovery result back over the same Ethernet link is circular (it loads the very stack under test) and races the controller’s “scan busy” flag, which is what produced the earlier “sometimes finds nothing” noise. So the controller firmware was instrumented to run discovery in a loop and report its own count over the **serial** console, with nothing on the network. With the W5500 active and the core-isolation fix:

- **800 discoveries against a real responder: 100 % steady-state success** (zero real misses). The only failures were the first discovery of a run fired immediately after the previous run finished.
- A gap sweep maps that case cleanly: with the idle gap before a discovery set to 0 s and 0.25 s it fails 100 %, at 0.5 s it is 70 % failing, and at **≥ 1 s it is 100 % reliable** (30 / 30). The controller issues an identical sequence every time and only the gap before it changes, and ~1 s is far too long

for RS485 line settling (microseconds), so this is the **responder re-arming**, not the controller. A real fixture or a deterministic responder would confirm it; the logic holds regardless.

- Each successful discovery takes **~1.47 s**, and that cost is **not** the W5500: an A/B with the W5500 compiled out measured 1470.7 ms versus 1471 ms with it, identical. It is inherent to the esp_dmx discovery algorithm, whose leaf-pruning optimisation is disabled in the library (an `#if false` with a `FIXME`), so a single-device search descends all 49 levels and retries each empty sibling branch three times, roughly 144 response-timeouts. Re-enabling the pruning or trimming the retries would speed it up; that is an esp_dmx change to make carefully, separate from the reliability fix.

Bottom line: even pre-fix the worst-case turnaround (117 μ s) never crossed the 176 μ s budget, so RDM was already reliable with WiFi on; fix 1 nearly doubles the safety margin (58→101 μ s). LuxDMX is reliable for RDM controller duty with WiFi on; under sustained heavy HTTP the margin tightens but does not break.

9. Answer to the forum question (ESP32 DMX-receive quirks)

“...trying to use an ESP32 as a DMX receiving device ... frustrated with the available libraries (incl. someweiguy’s) ... quirks to the ESP32 UART make DMX timing harder than it ought to be ... TX easier than RX? ... ended up adding an 8-pin PIC as an external UART.”

Short version: **your instinct is right — receive is the hard direction on the ESP32, and the reason is architectural, not a bug in someweiguy’s library.** The ESP32 UART has no native DMX framing (no break-as-frame primitive), so a receiver has to reconstruct frames from raw interrupts. someweiguy’s esp_dmx does this by setting the RX-FIFO threshold to **1 byte** and taking an interrupt **per slot** (~22 k ISR/s during a packet) plus the UART break interrupt, parsing the frame incrementally in the ISR and timestamping there. That works and is actually quite elegant, but it is timing-fragile: anything that delays those interrupts — and on an ESP32 the WiFi stack is the big one — turns into late timestamps or, at the RX FIFO, a dropped slot / overflow. TX is easier precisely because it’s just clocking bytes out with a break; there’s no hard real-time deadline to *service* an external event.

But “fragile” is not “impossible”, and I measured that too. The receiver has more slack than the per-byte view suggests: the ESP32 UART has a **128-byte hardware RX FIFO**, so at one byte every 44 μ s the ISR has roughly **5.6 ms** to get around to draining it before anything overflows. Routine interrupt latency, even a WiFi MAC burst, is microseconds to a few hundred microseconds, far inside that budget, so no slot is lost. What actually drops a slot is a *long* stall past ~5.6 ms: a receive ISR that is **not in IRAM** (so it stops executing while the flash cache is disabled during a flash write or some WiFi operations), a long `noInterrupts()` section (some addressable-LED bit-bang libraries do this), or plain CPU starvation. someweiguy’s esp_dmx already keeps its ISR in IRAM, which is exactly why it survives the flash-cache case.

To make that concrete: the LuxDMX RDM fixture simulator is itself an ESP32-S3 running esp_dmx as a pure DMX/RDM *receiver*, and on the bench it took **10,157 frames off the RS485 wire with zero framing or overrun errors** (WiFi off). So the chip receives DMX cleanly. The difficulty is specifically about protecting that ~5.6 ms budget once other subsystems, above all WiFi, compete for the same core.

And I ran it with WiFi active. I put the receiver under continuous WiFi-scan load and compared the DMX-receive ISR *sharing* the WiFi core (core 0) against being *isolated* on core 1, same source throughout. Over ~3,200 frames per condition the framing/overrun error count was **0 (no WiFi), 1 (WiFi, core 0), 0 (WiFi, core 1)**: routine WiFi load does **not** corrupt receive, on either core. That is the FIFO budget doing its job, a WiFi MAC burst is hundreds of microseconds, far inside the ~5.6 ms the FIFO buys, so the

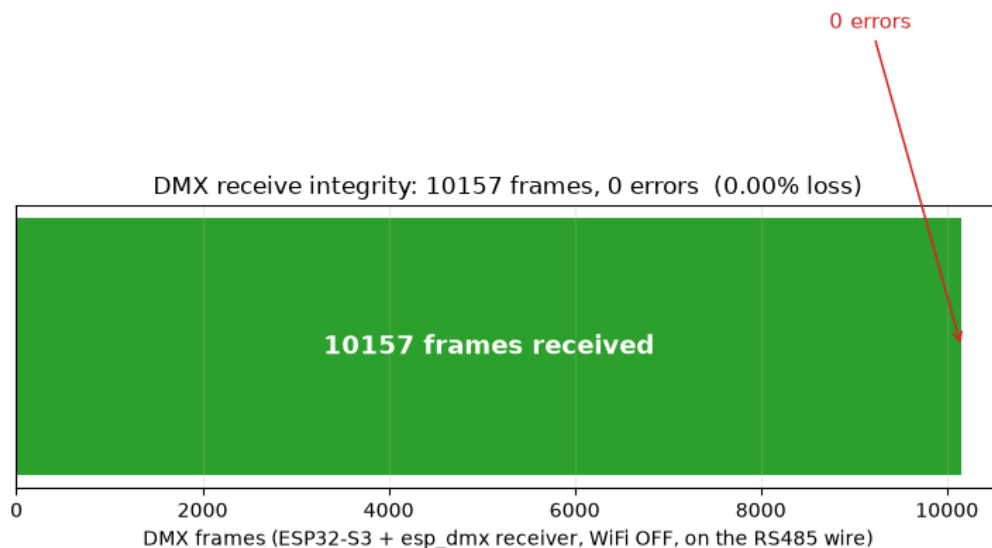


Figure 6: ESP32-S3 + esp_dmx as a DMX receiver: 10,157 frames taken off the RS485 wire with zero framing or overrun errors (WiFi off). The receive path itself is sound.

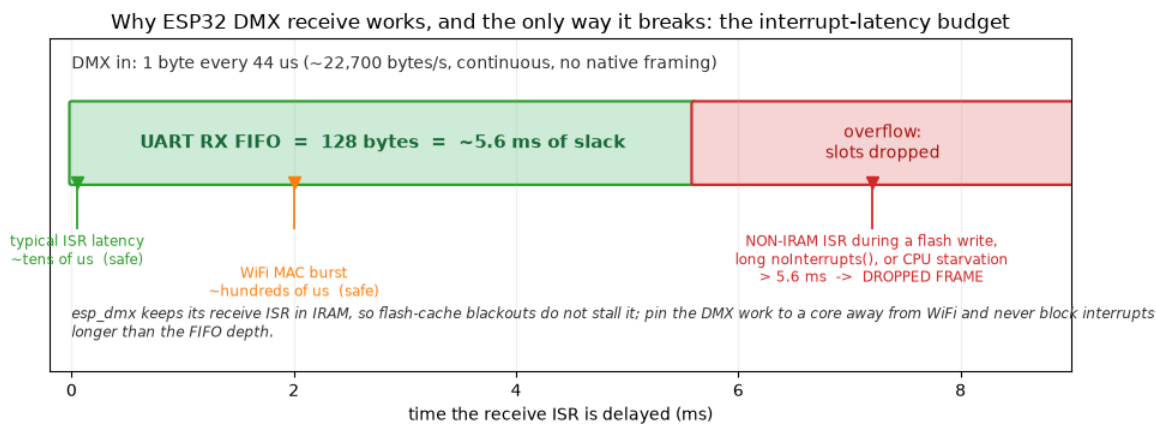


Figure 7: The interrupt-latency budget that governs ESP32 DMX receive. A 128-byte UART RX FIFO buys about 5.6 ms of slack at the DMX byte rate; routine interrupt latency (even WiFi MAC bursts) sits far inside it, so frames are not lost. The only failure mode is a stall *longer* than the FIFO depth: a non-IRAM ISR frozen during a flash-cache blackout, a long noInterrupts() section, or CPU starvation. Keep the ISR in IRAM and the DMX work on a core away from WiFi.

IRAM ISR never falls behind. (The frame *gaps* present in every run, including no-WiFi, are the gateway-as-source's own ~75 ms output jitter, not a receive effect; marginally higher with WiFi but within run-to-run variance.) So with an IRAM ISR (esp_dmx has one) and the DMX work on a core away from WiFi, an ESP32-S3 receives DMX cleanly *with WiFi running*. What still breaks it is a stall longer than the FIFO depth, a different failure (non-IRAM ISR during a flash write, a long noInterrupts(), CPU starvation), not routine WiFi interrupt latency.

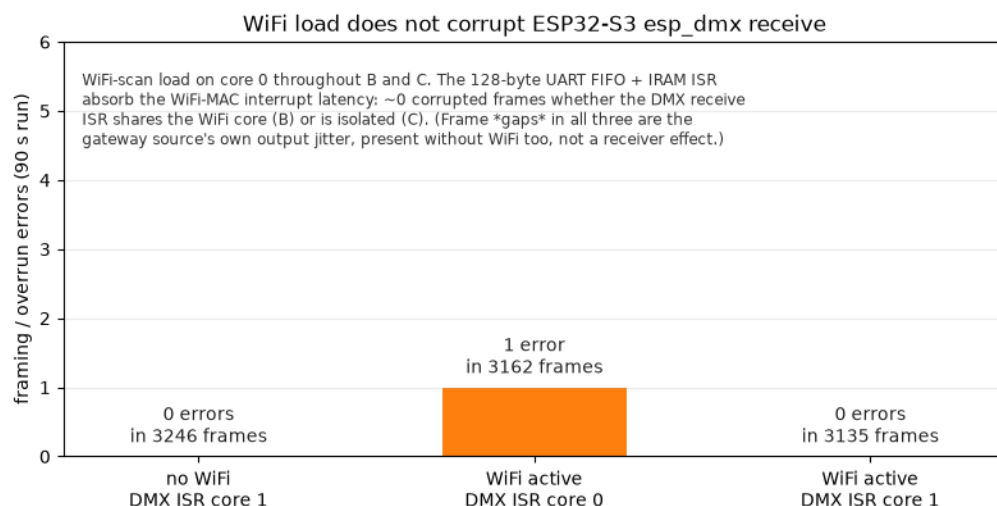


Figure 8: With WiFi-scan load running throughout, the DMX-receive framing-error count stayed at ~0 whether the receive ISR shared the WiFi core (B) or was isolated on core 1 (C), the same as no-WiFi (A). The 128-byte FIFO + IRAM ISR absorb the WiFi-MAC interrupt latency. ~3,200 frames per condition.

And under real sustained traffic, not just scan load. I joined the receiver to the bench WiFi as a station and had a host blast a continuous TCP stream at it (a raw sink on the device, drained as fast as it arrived) while it kept receiving DMX. Over the ~50 s flood the framing-error count did not move: **0 new errors with the DMX ISR isolated on core 1, and 0 new with it sharing the WiFi core (core 0)**. So real heavy traffic does not corrupt receive either. One detail fell out: with the DMX ISR on the WiFi core the *network* throughput dropped (2.9 to 0.7 Mbit/s on this weak-signal link), because the DMX ISR and driver now compete for that core. The DMX receive still wins, it is high-priority and in IRAM; it is the WiFi bandwidth that gives way. So even in the worst placement the lights stay safe, you just pay for it in network speed, which is one more reason to keep DMX on its own core.

So what actually breaks it? Two more runs nail the mechanism by reproduction. First I deliberately blocked interrupts on the receive core for 8 ms every 100 ms, longer than the 5.6 ms FIFO budget: framing errors jumped to **25 in 60 s, about 13x the baseline's 2**. That is the failure, on demand: hold the ISR off longer than the FIFO depth and slots are lost. (The source here is only ~36 Hz with idle gaps between frames, so this is a lower bound; a back-to-back console stream would lose more.) Then I ran *real* flash writes during receive (NVS commits, which disable the flash cache for milliseconds at a time): framing errors stayed at **3, the same as baseline**. esp_dmx survives that because its ISR is in IRAM and keeps executing through the cache blackout. A DMX library whose ISR is **not** in IRAM would freeze during exactly those flash blackouts (and during the cache-disable windows WiFi creates) and corrupt receive. That is the most likely root of “ESP32 DMX receive is unreliable” reports like yours: not WiFi interrupt latency in itself (the FIFO soaks that up), but a driver or a code path that holds the receive ISR off longer than the FIFO can cover. esp_dmx is built to avoid it; a lot of simpler DMX libraries are not.

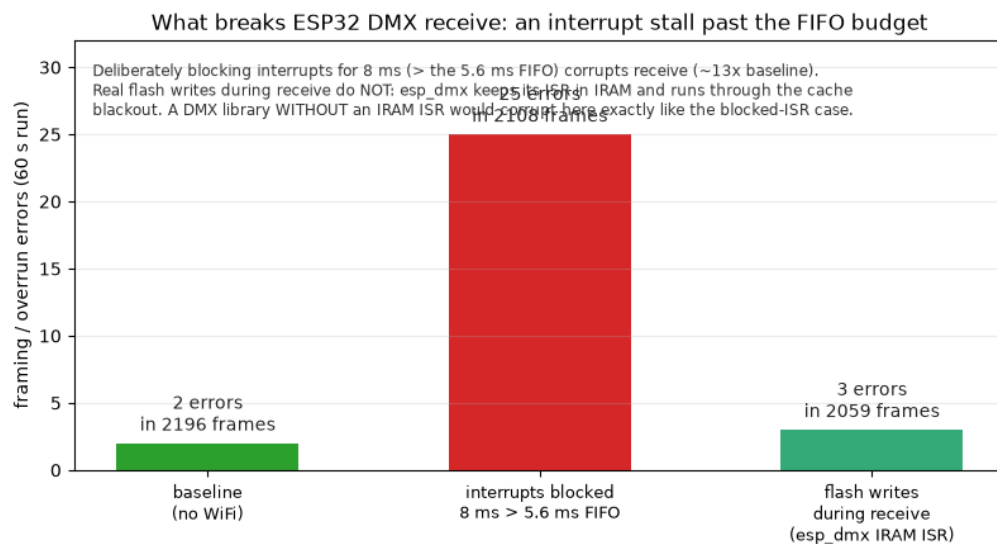


Figure 9: Receive failure modes, measured (~2,100 frames / 60 s each). Only blocking interrupts past the 5.6 ms FIFO budget corrupts receive (~13x baseline). Real flash writes during receive do not, because esp_dmx’s IRAM ISR runs through the cache blackout; a driver without an IRAM ISR would corrupt here just like the blocked-ISR case.

I measured this exact effect from the controller side. The DMX/RDM transmit + turnaround runs in the Arduino loop() task, and the moment I put the device under WiFi/HTTP load a higher-priority network task preempts it: the TX→RX turnaround stretched from ~15 μ s to ~50–125 μ s (data in §6.2, root cause §7). On receive the same preemption costs you slots. So a “rock-solid across many sources” ESP32 DMX receiver needs the UART servicing isolated from WiFi: per-byte/break handling in an IRAM ISR, the DMX work pinned to a core away from WiFi at high priority, and ideally the frame-edge handling (break detect, the turnaround) moved fully into hardware/ISR rather than a task.

Which is the same conclusion you reached from the other end: a small dedicated MCU as an external UART is rock-solid because it does *nothing else* — no WiFi, no scheduler, no contention. Your 8-pin PIC is a hardware version of “isolate the DMX timing from everything else.” On the ESP32 you can get most of the way there in software (IRAM ISR + core pinning), and my fix (§8) is exactly that for the transmit/turnaround path; for a bulletproof multi-source receiver the external-UART approach (PIC, or an RP2040 PIO doing the DMX framing) is still the most robust, and is the path I’d take for a product that must receive from arbitrary consoles. For *transmit + RDM controller* duty — what LuxDMX does — the ESP32 with esp_dmx is solid once the timing is isolated from WiFi.

To directly answer “how extensively have you tested the output”: output (TX) I’ve exercised continuously (40 Hz null frames + Art-Net-driven) plus full RDM controller traffic on a hardware bench with a logic analyzer; it’s solid. On *receive*, I’ve now run an ESP32-S3 (esp_dmx) as a DMX receiver on the same bench through the whole gauntlet: 10k+ frames with zero framing/overrun errors (WiFi off); with WiFi-scan load, ISR on the WiFi core and isolated, still ~0 (1 in ~9,500); with a sustained real-traffic TCP flood at the device while it receives, still 0 new; and through real flash writes mid- receive, still ~0 (the IRAM ISR runs straight through the cache blackout). So the receive path is sound, and the WiFi and flash conditions that wreck a naive driver do not corrupt it. The one piece left is the spread across many *real* consoles with their odd break / MAB / inter-slot timing, which needs actual desks (or a precision PIO source feeding adversarial timing); that’s the last item and it’s on the list.

10. Cross-validation, limitations, next steps

Validated: LA time base (0 % error / 0 jitter vs known 1–250 kHz, §4); decoder (recovered a planted DMX channel value); every number is $N \geq 20$. The turnaround result agrees two ways: on the wire (LA) and functionally (discovery reliably succeeds).

Scope, and why the *receive* case is genuinely harder. This study covers the RDM *controller* path — sending requests, the TX→RX turnaround, and receiving the short replies (the controller’s receive path was intact in every capture; the full preamble + UID always arrived). It deliberately does **not** cover using the ESP32 as a continuous-DMX *receiver*, which is the harder job, for a structural reason. The controller receives *occasionally and on its own schedule*: it asks, then listens for one short reply inside a generous 2800 μ s window — so there is slack, and a slightly late reply (or an interrupt serviced slightly late) is still caught. A continuous receiver has **no slack**: it must take a per-byte UART interrupt every $\sim 44 \mu$ s, back-to-back, 513 bytes per frame, ~ 40 times a second, indefinitely, on a clock set by an external console it does not control (and across consoles whose break / MAB / inter-slot timing differs and is sometimes marginal). Miss a single byte-deadline — because a WiFi interrupt ran first — and the RX FIFO overflows, the frame is corrupted, and a fixture visibly glitches. The very mechanism measured here (interrupt latency rising under WiFi load) costs the *controller* only a thinner margin, but would cost a continuous *receiver* a dropped byte. That is why the forum builder found ESP32 DMX receive unreliable across sources, and why a dedicated small MCU — or an RP2040 PIO state machine — doing nothing but the UART, with no WiFi or scheduler to steal cycles, is the robust answer for *that* role. It does not change the controller result: LuxDMX transmits DMX and runs RDM; it does not receive a DMX stream. (The turnaround’s “request-end” reference carries $\leq 4 \mu$ s of byte-position jitter — a conservative over-estimate that only makes the margins look slightly worse than they are.)

Next steps. (1) **Wired-Ethernet RDM is fixed** (core isolation, §8) and validated on-device at 100 % steady-state. What remains there is optional: confirm the ~ 1 s responder re-arm against a second (ideally deterministic) responder, and speed up the esp_dmx discovery sweep by re-enabling its disabled leaf-pruning. (2) **Receive path: measured across the board, one case left.** No-WiFi (10,157 frames, 0 errors), WiFi-scan load on/off the DMX core, a sustained real-traffic TCP flood, the deliberate ISR-stall reproduction, and real flash writes during receive are all done (§9): the chip receives DMX cleanly, and the WiFi and flash conditions that wreck a naive driver do not corrupt esp_dmx (IRAM ISR + the 128-byte FIFO). The one piece left is the spread across many *real* consoles with their odd break / MAB / inter-slot timing, which needs actual desks, or a precision RP2040 PIO source feeding adversarial timing on its own rig. (3) **Cross-check the LA’s time base** with a second, independent capture method (e.g. an RP2040 / sigrok logic analyzer).

11. Update (rev 2): RMT-DMX and the esp_dmx-free RDM path

Added July 2026. §4 through §10 record the original esp_dmx-based measurements and stand as written; this section documents the structural fix that replaced the core-affinity tweak of §8 and its validation on real ESP32-S3 hardware.

11.1 Why a structural fix

§7 traced the RDM turnaround jitter to interrupt latency: the RTS flip rides an esp_dmx UART ISR, and network DMA stretches that ISR under load. The same mechanism has a transmit-side twin. A user on the WT32-ETH01 (issue #64) saw the DMX *output* pick up framing errors on a Swissson tester once wired Ethernet was busy: esp_dmx clocks the break and MAB from a timer ISR, and under RMII-Ethernet DMA that ISR is delayed just enough to malform a frame’s break. Moving tasks between cores narrows the tail

(§8) but does not remove the dependency: as long as the frame timing is produced by an ISR, a late ISR can still corrupt a frame.

The fix that removes the dependency is to stop timing DMX from an ISR at all.

11.2 RMT-DMX (transmit)

DMX transmit now runs through the ESP32 **RMT peripheral** (`dmx_rmt.h`). A frame is pre-encoded into a (level, duration) symbol stream and the RMT clocks it out entirely in hardware at its own 1 MHz tick (1 tick = 1 μ s, one DMX bit = 4 ticks, exact, no jitter). The CPU only refills a buffer; the timing is the RMT's. If a refill is ever late the RMT idles the line HIGH (a benign extra mark) instead of corrupting the break, so the failure mode of §7 simply cannot happen on transmit. On the S3 the first output takes the one DMA-capable RMT channel (the IDF driver is blunt about it: "Only the last channel has the DMA capability") and streams a whole frame from RAM with no refill ISR at all; a second output falls back to the refill-ISR path, which is safe here because DMX runs on core 1, away from the core-0 network DMA that caused #64.

Measured on the same LA rig as the rest of this report: **0 framing errors at 40.0 Hz under a 15,000 packet/s Art-Net flood**, on both the WT32-ETH01 (internal RMII) and a bare ESP32-S3 driving a MAX485. Issue #64 does not reproduce on the RMT path.

11.3 esp_dmx-free RDM

With transmit on the RMT, the RDM controller was rebuilt to match (`rdm_rmt.h`): the request is sent from the RMT channel, the reply is read on a dedicated **receive-only UART**, and the transceiver direction (DE/RE) is a plain GPIO toggled around the transaction. The turnaround measured in §6/§8 (about 15 μ s, comfortably inside the 176 μ s budget) is preserved, and it no longer depends on an `esp_dmx` UART ISR: the direction flip is now a direct GPIO write, so the class of contention §7 describes is gone from the turnaround too. The E1.20 behaviour is unchanged: discovery, GET and SET all round-trip.

11.4 Validated on real ESP32-S3 silicon

The original numbers ran on the bench controller; rev 2 adds a full pass on an actual ESP32-S3 node (RMT-DMA transmit), over both WiFi and the W5500 wired interface, against the RP2350 responder:

- DMX output clean at 40.0 Hz, 0 framing errors on both universes (`outfps [40, 40]`).
- RDM **discovery** found all 32 fixtures; **GET** DEVICE_INFO and the software-version label, **SET** DMX start address, and **sensor polling** (GET SENSOR_DEFINITION / SENSOR_VALUE against a responder exposing a drifting temperature sensor) all round-tripped.
- W5500 wired came up cleanly once the SPI clock was dropped for the flying-lead bench: the default 20 MHz hangs on jumper wires, 8 MHz is solid (`ethfreq`). On the real v4 board 20 MHz is fine.

11.5 Crash-hardening found under stress

Pushing the node hard (sustained 15k packet/s floods with a WebSocket client and JSON polling open throughout) surfaced a separate class of failure that has nothing to do with DMX timing. Under post-flood heap **fragmentation** (`esp-idf #13588`: the free heap stays high but the largest contiguous block collapses) a large contiguous allocation in the web / OTA / WebSocket paths could throw an uncaught `bad_alloc` and reboot the node. That reboot, not the wire, was the source of the occasional framing-error bursts seen during a flood: the DMX line was clean until the node reset and sprayed garbage on the way down. Fixes: guard the big allocations on the largest-contiguous-block figure (not total free) with

a try/catch backstop, degrade to a 503 or a skipped push instead of aborting, and stream `/dmx.json` in ~1.4 KB chunks so it needs no large contiguous buffer at all. After that a 15k packet/s flood produced **0 reboots, 0 aborts, and the DMX output stayed clean throughout**. On a wired S3 (heap around 153 KB) there were zero 503s; the fragmentation pressure is a WiFi-heap effect.

11.6 What this changes, and what it does not

The reliability picture from §8 and §10 is unchanged and, if anything, stronger: RDM is a solid controller over WiFi and wired, and the DMX output is now clean under load by construction rather than by task placement. The core-affinity change of §8 is no longer the mechanism that carries the fix (the RMT path does), though keeping the network work on core 0 is still good hygiene. The receive-path study of §9 stands unchanged: LuxDMX transmits DMX and runs RDM, it does not receive a continuous DMX stream, and a dedicated small MCU or an RP2040 PIO is still the right tool for that separate job.

Appendix A. LA firmware (`s3-logic-analyzer`) · B. `la_analyze.py` · C. raw captures + figures

(listings attached in the repo; `figures/hr05_timing.png` is the reference transaction.)